SWAN DIGITAL SOLUTIONS

FULL STACK DEVELOPER — TECHNICAL ASSIGNMENT

Project: Lead Management System

Position	Full Stack Developer
Estimated time	2–3 days
Submission	Source code + deployed application
Technology	Candidate may choose the stack

Objective

Build a small, production-style Lead Management System for a software/digital marketing company. The goal is to evaluate your ability to design, develop, integrate, and deploy a complete full-stack application.

1. Authentication

• User registration
• User login and logout
• Protected dashboard and API routes
• Secure password storage

2. Lead Management

Users should be able to create, view, edit, and delete leads.

Lead fields: Name, Email, Phone Number, Company, Service Interested, Status, Follow-up Date, Notes, Created Date.

Status: New, Contacted, Qualified, Converted, Lost.

3. Dashboard

Display Total Leads, New Leads, Contacted Leads, Qualified Leads, Converted Leads, and Lost Leads. Include at least one meaningful chart/visualization.

4. Search & Filtering

• Search by name/email/phone
• Filter by lead status
• Filter by date
• Pagination

5. Backend & API

Create a proper backend API with authentication endpoints, lead CRUD endpoints, input validation, appropriate HTTP status codes, error handling, protected endpoints, and secure handling of sensitive information.

6. Database

Use a proper database to persist application data. Design the database structure appropriately.

7. UI/UX

The application should be clean, professional, responsive, easy to navigate, and usable on desktop and mobile. Functionality and code quality are equally important.

8. Deployment

Deploy the complete application for online testing. Submit the live frontend URL, live backend/API URL if applicable, database/setup documentation, and GitHub repository.

9. README

Include project overview, tech stack, features, setup instructions, environment variables required, API endpoint list/documentation, database information, deployment details, and demo credentials. Never commit secrets, passwords, API keys, or .env files.

Bonus Features

Optional: Admin/Sales roles, CSV export, lead activity/history, follow-up reminders, Docker, automated tests, API documentation, strong loading/error/empty states, and excellent mobile responsiveness.

Submission Requirements

1. GitHub repository
2. Live deployed URL
3. Demo login credentials
4. README.md
5. Any additional documentation you feel is useful.

Evaluation Criteria

Functionality • Frontend quality • Backend/API design • Database design • Authentication & security • Code quality • Error handling & validation • Responsive UI • Deployment • Problem-solving and overall implementation.

Important

You are free to choose the technology stack and architecture. We are more interested in how you design and implement the solution than in using a particular framework. The project should be your own work. Documentation, libraries, and AI tools may be used as development assistance, but you should be able to explain your code, architecture, and technical decisions during evaluation.